

RESEARCH ARTICLE

Accurate Detection of Solar Panel Defects Using RMN-YOLO With Multi-Scale Edge and Attention Enhancements

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ABSTRACT Accurate and efficient detection of defects in solar panels is essential for maintaining the stability and reliability of photovoltaic (PV) power generation systems. However, small defect sizes, complex backgrounds, and varying illumination conditions pose significant challenges to conventional detection methods. To address these issues, this study proposes RMN-YOLO, an enhanced detection model based on YOLOv11—the latest real-time object detection framework that integrates multi-scale feature extraction and high-speed inference. RMN-YOLO introduces three key improvements: a Residual Spatial and Channel Reconstruction Convolution (RSCConv) module in the backbone to strengthen spatial-channel feature representation; a Multi-Scale Edge Information Enhancement (MSEE) module in the neck to capture fine-grained defect boundaries; and a Dynamic Head (DyHead) with multi-attention mechanisms to improve small-object recognition and contextual awareness. Experiments on a publicly available solar panel defect dataset demonstrate that RMN-YOLO achieves a precision of 93.3%, recall of 82.7%, and mAP@50 of 90.5%, surpassing mainstream detectors such as SSD, Faster R-CNN, RetinaNet, and multiple YOLO variants. These results verify that RMN-YOLO effectively addresses the challenges of small-target detection and complex backgrounds, offering a practical and reliable solution for intelligent PV defect inspection.

INDEX TERMS Solar panel defect detection, YOLOv11, RSCConv, multi-scale features.

I. INTRODUCTION

With the rapid development of renewable energy technologies, solar power has become one of the most promising sources for sustainable electricity generation. As the core component of photovoltaic (PV) systems, solar panels directly determine the overall energy conversion efficiency and operational stability. However, long-term outdoor exposure often leads to various defects such as micro-cracks, hot spots, delamination, foreign object occlusion, and burn marks [1], [2]. These issues not only reduce power output but may also cause severe safety risks, including overheating and fire hazards. Therefore, accurate and timely defect detection is essential to ensure the reliability and economic performance of PV systems [3].

The associate editor coordinating the review of this manuscript and approving it for publication was Chao Tong².

From the perspective of clean energy promotion and environmental protection, timely and accurate detection and identification of solar panel defects are crucial for ensuring the efficient and stable operation of solar systems [4]. If defects are not detected and addressed at an early stage, they may not only degrade the performance of individual modules but also trigger cascading effects across the entire PV array, resulting in greater energy losses, higher maintenance costs, and ultimately undermining the economic viability and social acceptance of solar energy applications [5], [6], [7].

Nevertheless, solar panel defect detection faces numerous challenges and technical difficulties. First, defects exhibit diverse types and complex morphologies, often presenting subtle features that are difficult to detect with the naked eye. Second, constrained by complex outdoor environmental conditions (e.g., intense sunlight, reflections, wind, and dust),

traditional manual inspection methods suffer from low efficiency, high costs, and inconsistent accuracy. Third, mainstream detection techniques such as infrared thermography and electroluminescence (EL) imaging, while effective in laboratory settings, encounter practical limitations in large-scale deployment and real-time monitoring scenarios, including poor operability, significant imaging noise, and cumbersome data processing [8]. Additionally, variations in materials and structures among different types of PV panels impose higher demands on the generalization capability of defect detection algorithms.

In recent years, deep learning-based computer vision techniques have significantly advanced defect detection accuracy and automation. While these studies have achieved remarkable progress, several challenges remain unresolved. Most existing models struggle with micro-defect recognition, insufficient edge feature extraction, and poor robustness in complex outdoor environments. Moreover, improvements in accuracy often come at the cost of increased computational complexity, limiting real-time deployment on embedded platforms. Therefore, the development of efficient, intelligent, and automated defect detection technologies for solar panels not only directly contributes to improving PV system performance and reducing operational costs but also provides robust technical support for advancing green energy initiatives and achieving carbon neutrality goals. This research direction has emerged as one of the key topics in the field of new energy technologies. To address these challenges, we propose an improved RMN-YOLO model based on YOLOv11. The main contributions of this study are as follows:

- (1) Residual Spatial and Channel Reconstruction Convolution (RSCConv) is introduced in the backbone network to enhance the model's ability to extract multi-scale and spatially diverse feature representations.
- (2) Multi-Scale Edge Information Enhancement (MSEE) is employed in the neck network, combining edge reinforcement mechanisms with multi-scale feature partitioning to strengthen the extraction of fine edge details in solar panel images, particularly for small targets.
- (3) To mitigate missed detections and false alarms caused by complex background interference, a Dynamic Detection Head (DyHead) with a multi-attention mechanism is adopted, improving the model's sensitivity to small target features.
- (4) Extensive experiments on a publicly available solar panel defect dataset demonstrate that RMN-YOLO achieves state-of-the-art accuracy and real-time performance compared with mainstream detection algorithms.

These contributions highlight RMN-YOLO's practical potential for intelligent photovoltaic defect inspection, offering a reliable and efficient foundation for future embedded and UAV-based inspection systems.

II. RELATED WORK

Currently, research on solar panel defect detection primarily focuses on two major directions: traditional image processing methods and intelligent algorithms based on deep learning. Traditional methods mainly rely on infrared thermography, electroluminescence (EL) imaging, and visible-light images, employing techniques such as edge detection, texture analysis, and threshold segmentation to identify defective regions [9], [10]. These approaches are simple to implement and highly interpretable, but they are susceptible to interference in complex backgrounds, low-contrast defects, or non-uniform lighting conditions, resulting in limited robustness. In recent years, with advancements in computational power and artificial intelligence, deep learning methods have gradually become a research hotspot. Models such as convolutional neural networks (CNNs) have been widely applied to the detection, segmentation, and classification of solar panel defects, significantly improving accuracy and automation levels [11]. Particularly in handling large-scale image data, extracting fine-grained features, and adapting to multiple defect types, deep learning has demonstrated superior performance. However, challenges remain, including the demand for large training datasets, poor model interpretability, and high reliance on computational resources in real-world deployment. Many researchers have made outstanding contributions in addressing these issues.

Zheng et al. developed a frequency-domain filtering-based surface defect detection algorithm that employs Fast Fourier Transform (FFT) to convert target images into spectral representations. The methodology implements a frequency-domain filtering strategy to eliminate high-frequency components and noise from the spectrum, followed by reconstruction of the spatial-domain image through Inverse Fast Fourier Transform (IFFT). Subsequently, the Otsu automatic threshold segmentation method is applied for regional partitioning, enabling precise identification and extraction of surface defects in solar panels [12].

Huang et al. addressed the low detection efficiency of traditional defect detection methods by improving the YOLOv5 algorithm. Specifically, they first enhanced the Coordinate Attention mechanism to obtain an attention mechanism with broader target coverage, which not only effectively captures feature information but also expands the perceptual range for target features. Additionally, they replaced the commonly used Coupled Head in YOLO series with a Decoupled Head, enabling more precise task-specific processing and thereby improving detection accuracy [13].

Cao et al. incorporated an attention mechanism and a bidirectional feature pyramid network to enhance defect detection accuracy, while introducing lightweight GhostConv modules and Gaussian Error Linear Unit (GELU) activation functions to reduce model parameters and improve inference speed. This algorithm demonstrated improvements in both precision and processing speed, showcasing its advantages for rapid and accurate photovoltaic defect detection [14].

Prabhakaran et al. proposed a Multivariate Deep Learning Model (RMVDM) for defect identification and localization in photovoltaic panels. This methodology incorporates diverse defect characteristics to enable comprehensive defect detection and precise localization. The approach employs a region-based histogram approximation algorithm for image preprocessing, followed by feature extraction using grayscale quantization algorithms. Subsequently, high-order texture localization techniques are implemented to accurately pinpoint defect locations [15].

While the aforementioned studies have achieved significant progress in solar panel defect detection by improving both accuracy and automation levels, they still face persistent challenges including difficulties in micro-defect recognition, insufficient edge information extraction, and false positives/negatives caused by complex background interference. To address these limitations, this study proposes an enhanced RMN-YOLO model based on YOLOv11, which systematically integrates three key innovations: the RSCConv module for multi-scale feature extraction, the MSEE edge enhancement architecture, and a DyHead [16] detection head incorporating multi-attention mechanisms. This integrated approach is specifically designed to (1) enhance the model's sensitivity to defects across various scales, (2) improve edge detail representation, and (3) strengthen detection robustness in complex scenarios.

III. METHODS

A. RMN-YOLO NETWORK MODEL

RMN-YOLO is an enhanced model for solar panel defect detection developed upon the YOLOv11 framework. As a state-of-the-art single-stage object detector, YOLOv11's architecture comprises three core components: a Backbone network, feature fusion Neck module, and detection Head. In general, the backbone network serves as the feature extractor that learns visual representations from input images; the neck is a feature fusion module that integrates multi-scale information from different layers; and the detection head is responsible for predicting object categories and bounding boxes based on the fused features. The Backbone employs an optimized Cross Stage Partial (CSP) network integrating C3K2 convolutional modules, SPPF pyramid pooling, and Positional Spatial Attention (PSA) mechanisms, enabling efficient multi-scale feature extraction. The Neck module utilizes an advanced feature pyramid design for deep integration of detailed and semantic features, while the Head maintains an optimal balance between detection speed and accuracy.

The RMN-YOLO network architecture diagram is shown in Figure 1. Building upon this foundation, RMN-YOLO incorporates three strategic enhancements: (1) The Backbone is augmented with RSCConv modules to strengthen multi-scale and spatial feature representation; (2) The Neck integrates MSEE modules that combine edge reinforcement with multi-scale feature partitioning for precise extraction of fine-grained edge characteristics in solar panel defects;

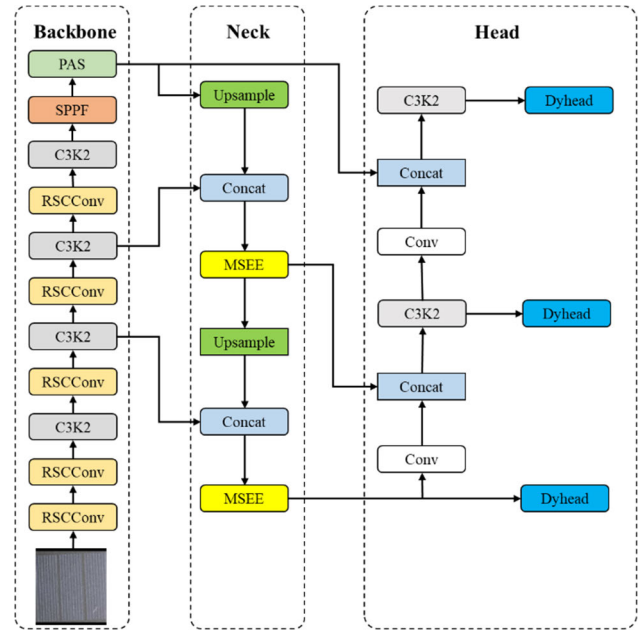


FIGURE 1. Structure diagram of RMN-YOLO.

(3) The detection Head adopts a DyHead structure with multi-attention mechanisms to significantly improve small defect detection capability in complex backgrounds. These innovations collectively enable RMN-YOLO to achieve substantially improved detection accuracy and robustness while maintaining high computational efficiency.

B. RESIDUAL SPATIAL AND CHANNEL RECONSTRUCTION CONVOLUTION

In backbone networks, while standard convolution operations can effectively extract local image features, they exhibit notable limitations when handling multi-scale targets and complex spatial distributions due to their restricted receptive fields. These operations struggle to simultaneously capture fine-grained details and global contextual information, particularly when processing small, fine-grained targets such as solar panel defects. Furthermore, standard convolutions process channel and spatial information independently, failing to adequately exploit their inherent interdependencies. To address these issues, this study introduces the RSCConv module into the backbone network. By incorporating residual connection mechanisms and jointly modeling spatial-channel dimensional relationships, the proposed module significantly enhances the model's capacity to perceive and represent multi-scale features with complex spatial distributions, thereby enabling more precise defect feature extraction. The RSCConv network architecture diagram is shown in Figure 2.

The RSCConv module comprises several key components: a Spatial Reconstruction Unit (SRU), a Channel Reconfiguration Unit (CRU), N Bottleneck blocks, three main-path convolutional layers (Conv), direct residual connections,

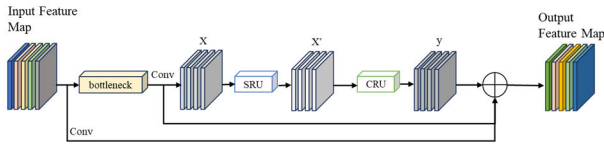


FIGURE 2. Structure diagram of RSCConv.

and cross-layer connections via Conv layers. The processing pipeline operates as follows: The input feature X first passes through a Bottleneck block and a 1×1 convolutional layer to extract preliminary features from the preceding layer. These features then undergo spatial refinement through the SRU, followed by channel refinement via the CRU to produce output feature Y . Finally, the module concatenates the outputs from two residual paths with Y to generate the final extracted features. In the Separate operation, the input feature X first undergoes feature map separation via Group Normalization (GN), followed by standardization through mean subtraction (μ) and division by the standard deviation (σ).

$$X_{out} = GN(X) = \gamma \frac{X - \mu}{\sqrt{\sigma^2 + a}} + \partial \quad (1)$$

Here, N denotes the batch size, C represents the number of channels, while H and W correspond to height and width respectively. The terms μ and σ denote the mean and standard deviation of X , where a is an adjustable error term, and γ and ∂ are adaptive affine transformation parameters. Subsequently, the importance of different feature maps is computed, and the weight values W are obtained through a Sigmoid function.

$$W_{\gamma} = \frac{\gamma_i}{\sum_{j=1}^C \gamma_j}, i, j = 1, 2, \dots, C \quad (2)$$

$$W = Gate(Sigmoid(W_{\gamma}(GN(X)))) \quad (3)$$

Finally, the Separate operation outputs x_1^w and x_2^w :

$$X_n^w = W_n \times X, n = 1, 2 \quad (4)$$

After entering the Reconstruction operation, cross reconstruction is used to fully combine the two features and enhance the information flow between them.

$$X_{11}^w \oplus X_{22}^w = X^{w1} \quad (5)$$

$$X_{21}^w \oplus X_{12}^w = X^{w2} \quad (6)$$

$$X^{w1} \cup X^{w2} = X^w \quad (7)$$

Among them, $\oplus$ represents element wise addition, and $\cup$ represents concatenation.

The CRU module comprises three components: Split, Transform, and Fuse. In the Split stage, the spatial features X^w output from the SRU are divided into two parts based on a predefined ratio a ($a \in [0, 1]$), where one part contains aC channels and the other contains $(1 - a)C$ channels. Subsequently, 1×1 convolutions are applied to compress the feature maps, producing the upper-channel feature map X_{up} and the lower-channel feature map X_{low} , thereby

enhancing computational efficiency. During the Transform stage, the upper-channel features are independently processed using Grouped Convolution (GWC) and Pointwise Convolution (PWC), and the results are summed to yield the final upper-channel feature map $Y1$. For the lower-channel features, a PWC operation is performed and the result is directly concatenated with X_{low} to generate the lower-channel feature map $Y2$.

$$Y1 = M^G X_{up} + M^{p1} X_{up} \quad (8)$$

$$Y2 = M^{p2} X_{low} \cup X_{low} \quad (9)$$

In the final merge operation, the channel information is first counted:

$$S_{1,2} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^H Y_c(i, j) \quad (10)$$

The features $S1$ and $S2$ are then stacked together, followed by Channel Attention (CA) mechanism to extract the feature importance vector $c1, 2$, ultimately merging the channels to output the final channel feature Y .

$$c1 = \frac{e^{s1}}{e^{s1} + e^{s2}} \quad (11)$$

$$c2 = \frac{e^{s2}}{e^{s1} + e^{s2}} \quad (12)$$

$$c1 + c2 = 1 \quad (13)$$

$$Y = c1 Y1 + c2 Y2 \quad (14)$$

In summary, RSCConv effectively reduces redundant features, significantly decreasing the model's computational load and parameter count, thereby lowering computational costs. Furthermore, it enhances feature representation capability, generating more discriminative and expressive features. Additionally, RSCConv offers plug-and-play functionality and strong generalization across various applications.

C. VARIABLE KERNEL CONVOLUTION MULTI-SCALE EDGE INFORMATION ENHANCEMENT

The Multi-Scale Edge Information Enhancement (MSEE) module is designed to strengthen the extraction of fine edge details of small targets in solar panel images through a synergistic integration of edge enhancement mechanisms and multi-scale feature partitioning. The Edge Enhancer mechanism is based on the principle of high-frequency information separation. It begins with low-frequency suppression: the input feature map is smoothed using average pooling to extract its low-frequency background information, denoted as Flow. Then, edge enhancement is performed by subtracting the low-frequency component Flow from the original feature map F_{in} , yielding the high-frequency edge information $F_{edge} = F_{in} - Flow$. Finally, edge feature enhancement is carried out by applying a convolution operation $Conv(F_{edge})$ to perform nonlinear transformation on the edge information, further amplifying the edge response. This result is then added back to the original feature to obtain the final output

$F_{out} = F_{in} + \text{Conv}(F_{edge})$, thereby enhancing edge details while preserving the original features. The structure of the Edge Enhancer module is illustrated in Figure 3.

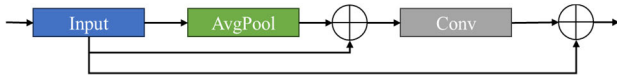


FIGURE 3. Edge enhancer module structure.

The MSEE module performs multi-scale edge extraction through adaptive average pooling, as illustrated in Figure 4. The process can be described as follows: the input image is first divided into multiple branches, each of which undergoes an adaptive average pooling operation with a different kernel size. This operation partitions the features into sub-regions with varying receptive fields, enabling the module to capture more local details. Subsequently, each branch passes through two convolutional layers sequentially to extract features. An upsampling operation is then applied to restore the resolution of each feature map to match that of the input. These upsampled features are fed into an edge enhancement module to obtain edge-enhanced outputs. The outputs from all branches are then concatenated along the channel dimension, forming a feature map that aggregates edge information from multiple scales. This concatenated feature map is further processed by an additional convolutional layer to enhance feature integration and optimization. The final output effectively combines edge information at different scales, making it particularly suitable for detecting small targets.

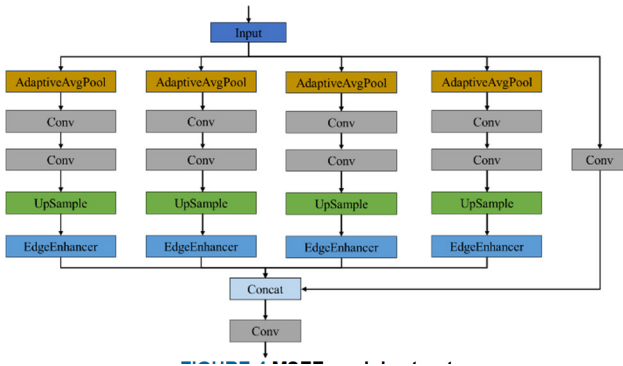


FIGURE 4. MSEE module structure.

In solar panel defect detection, MSEE module plays a critical role in improving the model's ability to capture fine-grained edge features of small and subtle defects. By leveraging adaptive average pooling across multiple scales, MSEE effectively extracts edge information from regions with varying receptive fields, enabling the model to detect localized defects that may be easily overlooked in complex or low-contrast backgrounds. The integrated edge enhancement mechanism further amplifies high-frequency details, allowing the network to emphasize defect contours and structural irregularities without sacrificing contextual information. As a result, the MSEE module significantly

enhances the precision and robustness of defect localization, particularly in challenging scenarios involving micro-cracks, scratches, or surface erosion on solar panels.

D. DYNAMIC HEAD MODULE

YOLOv11 demonstrates outstanding performance in terms of both detection accuracy and speed; however, its original detection head still presents certain limitations when it comes to detecting small objects. This issue is particularly evident in solar panel defect detection tasks, where the available feature information is often sparse and the distinction between adjacent objects is challenging, resulting in insufficient or lost information during feature extraction and recognition, which in turn degrades detection accuracy. Moreover, the original detection head in YOLOv11 does not fully incorporate contextual information during prediction, lacking a global perspective.

To address these issues, this study adopts the Dynamic Head (DyHead) to replace the original detection head in YOLOv11. DyHead introduces a unified attention-based structure that enhances object detection through three key mechanisms: level-wise feature integration, which performs scale-aware fusion of multi-level features based on semantic importance to improve detection of objects at different sizes; spatial-wise enhancement, which applies spatial attention modules to strengthen the model's perception of object locations; and channel-wise task adaptation, which enhances the model's ability to handle different prediction tasks through adaptive channel modulation. This unified detection head architecture effectively overcomes the limitations of traditional detection heads, which typically address only one of these challenges. The structure of DyHead is illustrated in Figure 5.

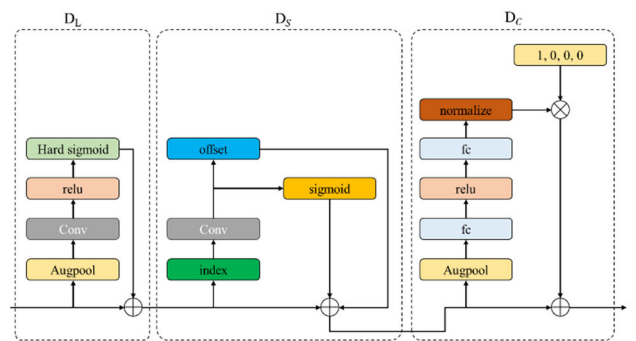


FIGURE 5. DyHead module structure.

Here, D_L , D_S , and D_C represent level-aware attention, spatial-aware attention, and channel-aware (task-aware) attention, respectively. For a given feature tensor $F \in \mathbb{R}^{L \times S \times C}$, the attention computation can be formulated as follows:

$$W(F) = D_C(D_S(D_L(F) \cdot F) \cdot F) \cdot F \quad (15)$$

Firstly, scale aware attention is introduced to dynamically fuse the semantic importance of features at different scales,

as shown in the following equation:

$$D_L(F) \cdot F = \sigma \left(f \left(\frac{1}{SC} \sum_{s,c} F \right) \right) \cdot F \quad (16)$$

$$\sigma(x) = \max \left(0, \min \left(1, \frac{x+1}{2} \right) \right) \quad (17)$$

Here, $f(\cdot)$ is a linear function, approximated using a 1×1 convolution, and σ denotes the hard-sigmoid activation function. Subsequently, a spatial position-aware attention module is introduced to help the model focus on and enhance its sensitivity to key spatial locations within the image. Specifically, deformable convolution is first employed to sparsify the attention learning process, followed by cross-level feature aggregation, as expressed in the following equation:

$$D_S(F) \cdot F = \frac{1}{L} \sum_{l=1}^L \sum_{k=1}^K \omega_{l,k} \times F(l; p_k + \Delta p_k; c) \times \Delta m_k \quad (18)$$

Here, K denotes the number of sparse sampling locations, $p_k + \Delta p_k$ represents the shifted position obtained by the learnable spatial offset Δp_k , which focuses on discriminative regions, and Δm_k is a learnable importance scalar at position p_k .

Finally, to enable joint learning and generalize different object representations, task-aware attention is introduced, which dynamically switches feature channels to support various tasks, as expressed in the following equation:

$$D_C(F) \cdot F = \max \left(\alpha^1(F) \cdot F_C + \beta^1(F), \alpha^2(F) \cdot F_C + \beta^2(F) \right) \quad (19)$$

Among them, $\theta(\cdot) = [\alpha^1, \beta^1, \alpha^2, \beta^2]T$ is used to control the activation threshold. Finally, apply the above three attention mechanisms in order and stack them in sequence.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. EXPERIMENTAL DATASET

The dataset used in this study is the publicly available solar panel defect detection dataset provided by PaddlePaddle. It comprises images collected at different time periods to ensure diversity in lighting conditions and defect appearances. The dataset is annotated with three types of defects: Crack, Grid, and Spot. A total of 2,400 images are included, with 1,800 samples used for training, 480 for validation, and 120 for testing. Figure 6 illustrates the target distribution within the dataset. As shown, the majority of defect instances in solar panel images are small in size, indicating that the detection task is predominantly focused on small object recognition. Moreover, the targets are primarily concentrated around the edges and center regions of the images, presenting additional challenges for accurate localization due to their dense and varied spatial distribution.

The dataset can be obtained from the following link:
<https://aistudio.baidu.com/datasetdetail/220416>.

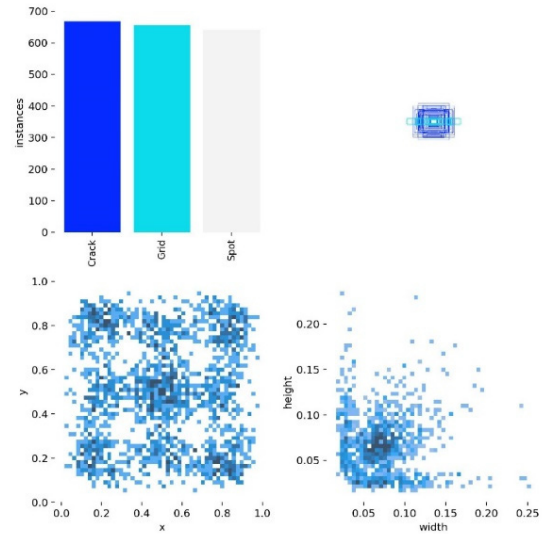


FIGURE 6. Label distribution.

B. EXPERIMENTAL ENVIRONMENT

The experiments were conducted on a system running Windows 10, with Python version 3.12 and the PyTorch 2.0 deep learning framework. CUDA version 11.8 was used for GPU acceleration. The hardware setup included an Intel(R) Core(TM) i5-12400 CPU, an NVIDIA GeForce RTX 4070 Super GPU with 12 GB of video memory, and 32 GB of RAM. During training, the input image size was set to 640×640 , the batch size was 16, the initial learning rate was 0.01, the number of worker threads was 8, and the SGD optimizer was applied.

C. EVALUATION METRICS

To demonstrate the effectiveness of the proposed algorithm, the selected evaluation metrics include Precision (P), Recall (R), mean Average Precision (mAP), number of parameters (Params), and frames per second (FPS). The mAP adopts the average precision mean when the IoU threshold is 0.5. A higher mAP indicates better model accuracy. Params reflect the model's lightweight nature, while FPS represents the number of frames processed per second—the higher the value, the faster the inference speed. The relevant calculation formulas are as follows:

$$P = \frac{TP}{TP + FP} \quad (20)$$

$$R = \frac{TP}{TP + FN} \quad (21)$$

$$mAP = \frac{1}{N} \sum_{i=1}^n \int_0^1 P(R) dR \quad (22)$$

In the formulas, TP (True Positive) indicates instances correctly predicted as positive; FP (False Positive) refers to cases where negative samples are mistakenly classified as positive; FN (False Negative) represents positive samples that are incorrectly predicted as negative.

TABLE 1. The ablation test results of RMN-YOLO.

Model	RSCConv	MSEE	DyHead	P/%	R/%	mAP/%	Params /M	FPS
A	×	×	×	89.1	79.8	87.4	9.4	90.3
B	√	×	×	90.9	81.5	88.9	9.5	87.7
C	√	√	×	91.6	81.4	89.1	9.3	91.3
D	√	√	√	93.3	82.7	90.5	9.6	89.6

D. ABLATION EXPERIMENTS

To evaluate the individual contributions of RSCConv, MSEE, and DyHead modules within the RMN-YOLO framework, a series of ablation experiments were conducted. These experiments systematically remove or isolate each component to assess its impact on overall model performance. By comparing detection results across different configurations—such as the baseline YOLOv11, RMN-YOLO without RSCConv, without MSEE, and without DyHead—it becomes possible to quantify the effectiveness of each module in enhancing feature extraction, edge detail preservation, and robustness to complex backgrounds. This analysis provides strong empirical evidence for the necessity and complementary nature of the proposed improvements in addressing the challenges of small target detection in solar panel defect images. The experimental results are shown in Table 1.

The results of the ablation study, as summarized in Table 1, clearly demonstrate the effectiveness of the proposed components—RSCConv, MSEE, and DyHead—within the RMN-YOLO framework. Model A, which serves as the baseline without any of the three modules, achieves a precision of 89.1%, a recall of 79.8%, and an mAP of 87.4%, with 9.4M parameters and an inference speed of 90.3 FPS. Upon integrating RSCConv alone (Model B), there is a noticeable improvement in all evaluation metrics, with precision increasing to 90.9% and mAP reaching 88.9%, indicating enhanced feature extraction capability, particularly in spatial and channel dimensions.

When both RSCConv and MSEE are employed (Model C), further gains are observed in precision (91.6%) and mAP (89.1%), while the recall remains relatively stable. This demonstrates that MSEE effectively enhances fine-grained edge information, contributing to more accurate localization of small defects without compromising computational efficiency, as evidenced by a slight reduction in parameter count (9.3M) and a high inference speed (91.3 FPS).

Finally, the full model (Model D), which incorporates RSCConv, MSEE, and DyHead, achieves the best overall performance, with precision, recall, and mAP rising to 93.3%, 82.7%, and 90.5%, respectively. The introduction of DyHead significantly improves contextual perception and small object sensitivity, leading to a substantial boost in recall and overall detection accuracy. Despite a slight increase in parameter count (9.6M) and a minor decrease in FPS (89.6), the model maintains real-time detection capability. These results collectively validate the complementary effects of the proposed modules and highlight the effectiveness of the RMN-YOLO

TABLE 2. Comparative experimental results of different convolutions.

Model	P/%	R/%	mAP/%	Params /M	FPS
Baseline	89.1	79.8	87.4	9.4	90.3
+RFAConv	90.5	81.2	88.8	11.7	82.6
+DWConv	87.3	80.0	86.2	7.8	102.5
+ODConv	90.4	80.3	87.9	10.2	86.4
+RSCConv	90.9	81.5	88.9	9.5	87.7

architecture in addressing the challenges of solar panel defect detection.

E. COMPARATIVE EXPERIMENTS OF CONVOLUTIONS

To verify the effectiveness of the proposed RSCConv module, a comparative experiment was designed to evaluate its performance against several representative convolutional structures, including RFAConv [17], DWConv [18], and ODConv [19]. By integrating each of these modules individually into the same baseline detection framework, we systematically compared their impact on key performance metrics such as precision, recall, mAP, model size, and inference speed. The comparative experimental results of different convolutions are shown in Table 2.

Table 2 presents the comparative results of different convolutional modules integrated into the baseline model, aiming to assess their impact on detection performance, model complexity, and inference speed. The baseline model, using standard convolution, achieves a precision of 89.1%, recall of 79.8%, and mAP of 87.4%, with 9.4 million parameters and a real-time speed of 90.3 FPS. Introducing RFAConv leads to notable improvements in both precision (90.5%) and recall (81.2%), with an increased mAP of 88.8%. However, this gain comes at the cost of higher computational complexity, as reflected by a significant parameter increase to 11.7M and reduced speed to 82.6 FPS, indicating a trade-off between accuracy and efficiency. The DWConv (Depthwise Separable Convolution) variant significantly reduces the parameter count to 7.8M and achieves the highest inference speed (102.5 FPS), demonstrating its lightweight nature. However, this efficiency gain results in decreased accuracy across all metrics, with precision dropping to 87.3% and mAP to 86.2%, suggesting it may not be suitable for tasks requiring high precision like defect detection.

ODConv achieves a balanced performance, improving precision (90.4%) and mAP (87.9%) over the baseline while moderately increasing model complexity (10.2M parameters) and maintaining acceptable speed (86.4 FPS). Although both ODConv and RSCConv aim to enhance the representational

capacity of convolutional kernels, their underlying mechanisms and effects differ significantly. ODConv dynamically generates convolutional weights along spatial, channel, and kernel dimensions, allowing it to adaptively adjust receptive fields. However, this flexibility incurs a high computational overhead due to the dynamic generation of multiple kernel weights and attention vectors, resulting in redundant computations when processing small or homogeneous targets. In contrast, RSCConv achieves adaptive feature reconstruction through a dual-dimensional refinement strategy rather than full dynamic kernel generation. Specifically, it employs a Spatial Reconstruction Unit (SRU) to refine local context and a Channel Reconfiguration Unit (CRU) to adaptively reweight channel dependencies via residual fusion. This design allows RSCConv to maintain a lightweight structure while effectively capturing spatial-channel correlations. Finally, RSCConv yields the best overall performance among the tested convolutional variants, with the highest precision (90.9%) and mAP (88.9%), and a recall of 81.5%. Notably, it maintains a lightweight structure (9.5M parameters) and near real-time speed (87.7 FPS), making it well-suited for practical deployment in solar panel defect detection. These results confirm the superior feature representation capability of RSCConv, particularly in capturing fine-grained defect characteristics.

F. COMPARATIVE EXPERIMENTS OF DIFFERENT MODELS

To validate the effectiveness of the proposed RMN-YOLO model, a series of comparative experiments were conducted against several mainstream deep learning-based object detection models, including SSD [20], Faster R-CNN [21], RetinaNet [22], CornerNet [23], as well as multiple versions of the YOLO series—YOLOv5s [24], YOLOv8s [25], YOLOv9s [26], YOLOv10s [27], and YOLOv11s [28]. These models represent both one-stage and two-stage detection paradigms and cover a wide range of accuracy-speed trade-offs. By benchmarking RMN-YOLO against these representative methods, the comparative study aims to comprehensively assess its performance in terms of precision, recall, detection accuracy (mAP) within the context of solar panel defect detection.

TABLE 3. Comparison of experimental results of different models.

Model	P/%	R/%	mAP/%
SSD	78.5	70.2	70.9
Faster R-CNN	72.3	68.5	63.8
RetinaNet	68.9	62.4	56.3
CornerNet	70.2	64.3	60.6
YOLOv5s	83.9	76.2	82.3
YOLOv8s	85.1	78.7	85.6
YOLOv9s	84.5	78.3	85.3
YOLOv10s	85.3	78.1	86.1
YOLOv11s	89.1	79.8	87.4
RMN-YOLO	93.3	82.7	90.5

As shown in Table 3, the proposed RMN-YOLO model outperforms all compared baseline methods across key

evaluation metrics, including precision, recall, and mAP, demonstrating its superior effectiveness for solar panel defect detection. Among the traditional two-stage detectors, Faster R-CNN, RetinaNet, and CornerNet perform relatively poorly, with mAP values of 63.8%, 56.3%, and 60.6%, respectively. These models, although capable of capturing complex features, suffer from lower efficiency and limited adaptability to small object detection, which is critical in solar panel defect tasks. SSD, as a one-stage model, achieves a better balance, with an mAP of 70.9%, but still falls short in terms of precision (78.5%) and recall (70.2%) when compared to modern YOLO variants. The YOLO family shows progressively improved performance with each version. YOLOv5s achieves an mAP of 82.3%, while YOLOv8s, YOLOv9s, and YOLOv10s demonstrate further gains, achieve progressive improvements, with YOLOv10s reaching a mAP of 86.1%. YOLOv11s, the latest version, achieves strong performance with a precision of 89.1%, recall of 79.8%, and mAP of 87.4%, which can be regarded as a strong reference baseline for comparison. However, RMN-YOLO, which integrates RSCConv, MSEE, and DyHead modules into the YOLOv11 architecture, surpasses all models with a precision of 93.3%, recall of 82.7%, and mAP of 90.5%. These results clearly demonstrate the effectiveness of the proposed enhancements in improving feature representation, edge detail extraction, and contextual awareness, particularly for small and subtle defects in complex solar panel imagery. RMN-YOLO's performance confirms its suitability for practical deployment in high-precision photovoltaic defect detection scenarios.

G. VISUALIZATION EXPERIMENT

To further evaluate the practical performance of RMN-YOLO in solar panel defect detection tasks, a series of visualization experiments were conducted in comparison with the baseline YOLOv11 model. The visualized experimental results are shown in Figure 7.

A total of six representative cases were selected, where each group presents side-by-side detection results—YOLOv11 on the left and RMN-YOLO on the right. The comparison highlights the enhanced detection accuracy, robustness, and detail perception of RMN-YOLO under varying conditions. In case (a), which involves the detection of a Spot-type defect, RMN-YOLO achieves a 7% higher precision compared to YOLOv11, demonstrating superior sensitivity to small and ambiguous targets. In case (b), despite the presence of strong sunlight that introduces visual interference, RMN-YOLO outperforms YOLOv11 in detecting Crack defects, showing better resilience to lighting variations. In case (c), RMN-YOLO successfully identifies an additional small Grid defect that YOLOv11 fails to detect, and it also yields higher detection confidence on the other two Grid instances, indicating improved capability in handling densely packed or low-contrast features. In case (d), YOLOv11 exhibits false duplicate detections for a single Crack, whereas RMN-YOLO provides a clean and accurate prediction with

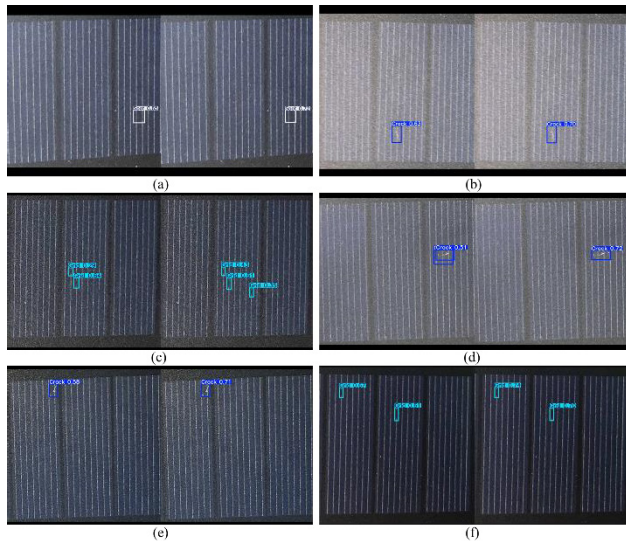


FIGURE 7. Visualized results of YOLOv11 and RMN-YOLO.

significantly higher precision, reflecting the effectiveness of the proposed attention and feature enhancement modules in suppressing redundant responses. In case (e), the small Crack defect is more accurately localized and delineated by RMN-YOLO than by YOLOv11, reaffirming its edge extraction and fine-grained feature discrimination advantages. Lastly, in case (f), where lighting conditions are relatively dim, RMN-YOLO maintains high detection precision on two Grid defects, whereas YOLOv11 suffers from reduced confidence and less accurate localization. Overall, these visual results clearly validate that RMN-YOLO not only improves detection accuracy across various defect types but also demonstrates stronger robustness in challenging imaging conditions such as glare, shadow, and low illumination. This further supports its applicability for real-world deployment in solar panel inspection scenarios.

V. CONCLUSION

In this study, we proposed RMN-YOLO, an enhanced solar panel defect detection model based on YOLOv11, integrating three key modules: RSCConv for enriched spatial and channel feature representation, MSEE for fine-grained multi-scale edge information extraction, and DyHead for improved contextual awareness and small object sensitivity. Extensive experiments on a public solar panel defect dataset demonstrate that RMN-YOLO significantly outperforms a series of state-of-the-art object detection models, including SSD, Faster R-CNN, RetinaNet, and various versions of the YOLO series. Ablation studies further validate the individual and combined contributions of the proposed modules, with consistent improvements in precision, recall, and mAP under real-time inference constraints. In addition, visual comparison experiments highlight RMN-YOLO's robustness in complex environments, such as strong illumination or dim lighting, and it demonstrates enhanced

capability in identifying subtle or overlapping defects, with improved confidence scores and reduced false detections. These results confirm the effectiveness and practical applicability of RMN-YOLO in real-world photovoltaic inspection scenarios, offering a reliable and efficient solution for automated solar panel defect detection. Although the current study primarily focuses on algorithmic design and validation in a controlled experimental environment, practical deployment on embedded platforms remains a promising direction for future research. In future work, we plan to further explore model compression, quantization, and hardware acceleration strategies to enable efficient deployment on embedded and resource-constrained platforms, thereby enhancing the real-world applicability of RMN-YOLO for autonomous photovoltaic inspection systems.

REFERENCES

- [1] S. Umar, M. S. Qureshi, and M. U. Nawaz, "Thermal imaging and AI in solar panel defect identification," *Int. J. Adv. Eng. Technol. Innov.*, vol. 1, no. 3, pp. 73–95, 2024.
- [2] J. A. Dhanraj, A. Mostafaeipour, K. Velmurugan, K. Techato, P. K. Chaurasiya, J. M. Solomon, A. Gopalan, and K. Phoungthong, "An effective evaluation on fault detection in solar panels," *Energies*, vol. 14, no. 22, p. 7770, Nov. 2021.
- [3] D. Dwivedi, K. V. S. M. Babu, P. K. Yemula, P. Chakraborty, and M. Pal, "Identification of surface defects on solar PV panels and wind turbine blades using attention based deep learning model," *Eng. Appl. Artif. Intell.*, vol. 131, May 2024, Art. no. 107836.
- [4] M. Meribout, V. K. Tiwari, J. P. P. Herrera, and A. N. M. A. Baobaid, "Solar panel inspection techniques and prospects," *Measurement*, vol. 209, Mar. 2023, Art. no. 112466.
- [5] U. Hijjawi, S. Lakshminarayana, T. Xu, G. P. M. Fierro, and M. Rahman, "A review of automated solar photovoltaic defect detection systems: Approaches, challenges, and future orientations," *Sol. Energy*, vol. 266, Dec. 2023, Art. no. 112186.
- [6] X. Chen, T. Karin, and A. Jain, "Automated defect identification in electroluminescence images of solar modules," *Sol. Energy*, vol. 242, pp. 20–29, Aug. 2022.
- [7] S. Jumaboev, D. Jurakuziev, and M. Lee, "Photovoltaics plant fault detection using deep learning techniques," *Remote Sens.*, vol. 14, no. 15, p. 3728, Aug. 2022.
- [8] A. Di Tommaso, A. Betti, G. Fontanelli, and B. Michelozzi, "A multi-stage model based on YOLOv3 for defect detection in PV panels based on IR and visible imaging by unmanned aerial vehicle," *Renew. Energy*, vol. 193, pp. 941–962, Jun. 2022.
- [9] J. Balzategui, L. Eciolaza, and D. Maestro-Watson, "Anomaly detection and automatic labeling for solar cell quality inspection based on generative adversarial network," *Sensors*, vol. 21, no. 13, p. 4361, Jun. 2021.
- [10] M. Waqar Akram, G. Li, Y. Jin, and X. Chen, "Failures of photovoltaic modules and their detection: A review," *Appl. Energy*, vol. 313, May 2022, Art. no. 118822.
- [11] L. Pratt, D. Govender, and R. Klein, "Defect detection and quantification in electroluminescence images of solar PV modules using U-net semantic segmentation," *Renew. Energy*, vol. 178, pp. 1211–1222, Nov. 2021.
- [12] J. Huang, K. Zeng, Z. Zhang, and W. Zhong, "Solar panel defect detection design based on YOLO v5 algorithm," *Heliyon*, vol. 9, no. 8, Aug. 2023, Art. no. e18826.
- [13] Y. Cao, D. Pang, Y. Yan, Y. Jiang, and C. Tian, "A photovoltaic surface defect detection method for building based on deep learning," *J. Building Eng.*, vol. 70, Jul. 2023, Art. no. 106375.
- [14] S. Prabhakaran, R. Annie Uthra, and J. Preetharoselyn, "Deep learning-based model for defect detection and localization on photovoltaic panels," *Comput. Syst. Sci. Eng.*, vol. 44, no. 3, pp. 2683–2700, 2023.
- [15] B. Zheng and Y. Yang, "Surface defect detection technology for solar panels based on machine vision," in *Proc. IEEE 7th Inf. Technol., Netw., Electron. Autom. Control Conf. (ITNEC)*, Sep. 2024, pp. 1831–1835.

- [16] X. Dai, Y. Chen, B. Xiao, D. Chen, M. Liu, L. Yuan, and L. Zhang, "Dynamic head: Unifying object detection heads with attentions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7369–7378.
- [17] X. Zhang, C. Liu, D. Yang, T. Song, Y. Ye, K. Li, and Y. Song, "RFACConv: Innovating spatial attention and standard convolutional operation," 2023, *arXiv:2304.03198*.
- [18] Z. Wang, X. He, Y. Li, and Q. Chuai, "EmbedFormer: Embedded depth-wise convolution layer for token mixing," *Sensors*, vol. 22, no. 24, p. 9854, Dec. 2022.
- [19] C. Li, A. Zhou, and A. Yao, "Omni-dimensional dynamic convolution," 2022, *arXiv:2209.07947*.
- [20] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot MultiBox detector," in *Proc. ECCV*, Oct. 2016, pp. 21–37.
- [21] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
- [22] H. Peng, Z. Li, Z. Zhou, and Y. Shao, "Weed detection in paddy field using an improved RetinaNet network," *Comput. Electron. Agricult.*, vol. 199, Aug. 2022, Art. no. 107179.
- [23] M. Nawaz, T. Nazir, M. Masood, A. Mehmood, R. Mahum, M. A. Khan, S. Kadry, and O. Thinnukool, "Analysis of brain MRI images using improved CornerNet approach," *Diagnostics*, vol. 11, no. 10, p. 1856, Oct. 2021.
- [24] Z. Wang, L. Jin, S. Wang, and H. Xu, "Apple stem/calyx real-time recognition using YOLO-v5 algorithm for fruit automatic loading system," *Postharvest Biol. Technol.*, vol. 185, Mar. 2022, Art. no. 111808.
- [25] H. Yi, B. Liu, B. Zhao, and E. Liu, "Small object detection algorithm based on improved YOLOv8 for remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1734–1747, 2024.
- [26] M. Yaseen, "What is YOLOv9: An in-depth exploration of the internal features of the next-generation object detector," 2024, *arXiv:2409.07813*.
- [27] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, and J. Han, "YOLOv10: Real-time end-to-end object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 107984–108011.
- [28] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.



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